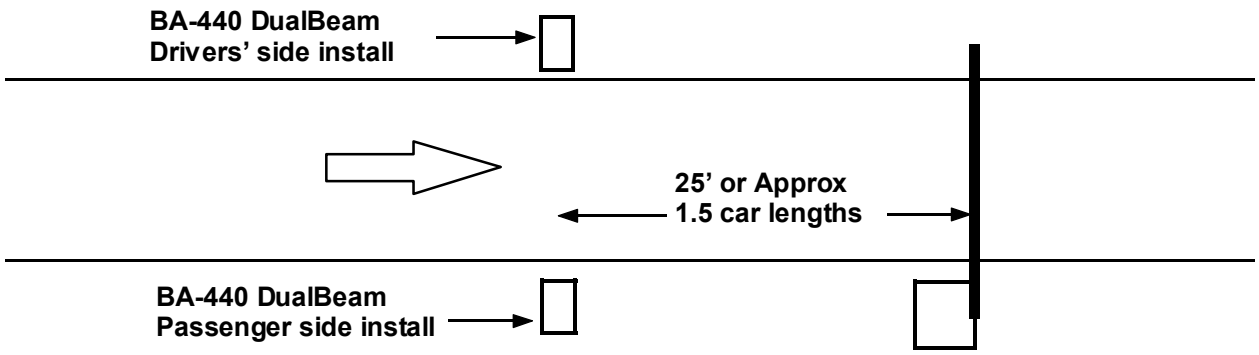
# BA-440 DualBeam Barcode Reader Quickstart Guide Ver 5

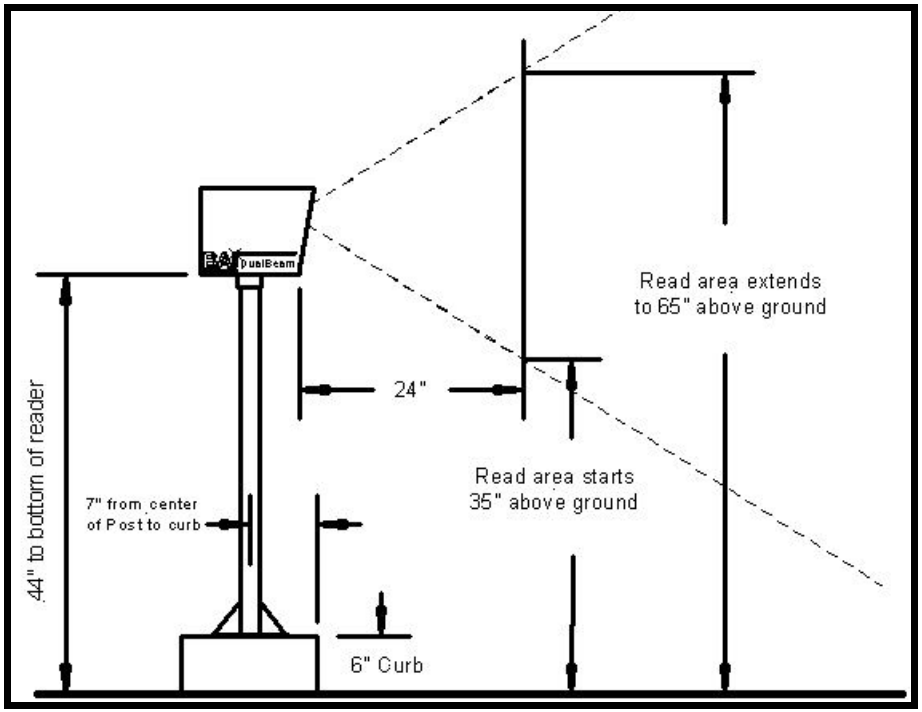



Overhead View

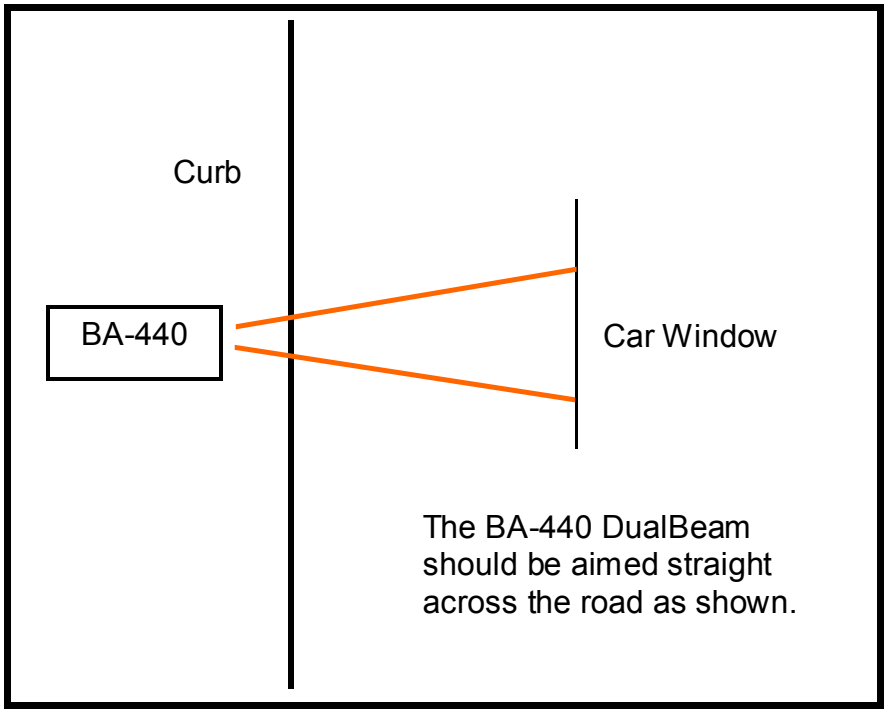
## 1. Determine where to place the BA-440 DualBeam

As shown in the figure above, the reader should be located about 1.5 car lengths (25') before the gate. It does not matter which side of the road it is mounted on, as long as the vehicle decals are on the same side. **Always place the reader and decals on the same side of the car**.

### Typical Installation Position



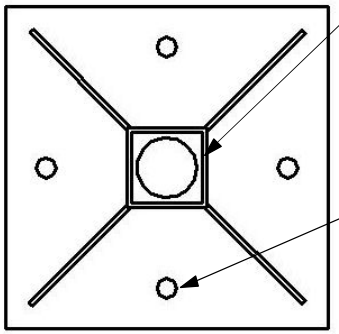

### Mounting Angle (overhead view)



# BA-440 DualBeam Barcode Reader Quickstart Guide Ver 5

## 2. Install mounting post

Mounting Post anchor bolts are not supplied. Use appropriate anchors for the material the post is anchored to. Conduit and/or wiring should be run up through the 3/4” diameter hole in the center of the mounting post.



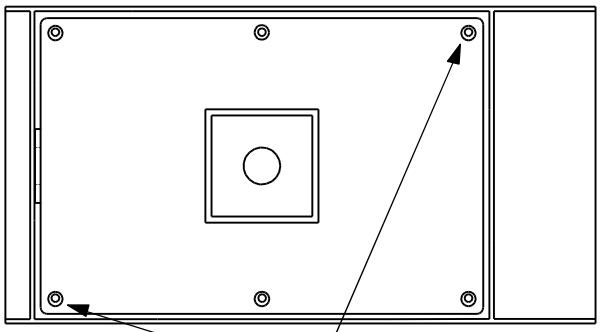

## 3. Recommended Tools

To assemble the BA-440 we recommend the following tools:

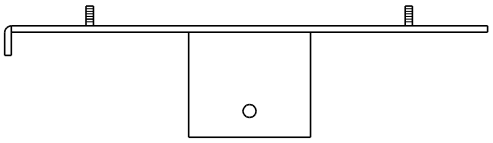
11/32” socket with ratchet
5/16” socket with extension
Recommend 10” long extension for sockets
Medium size Phillips screwdriver
Small flat screwdriver with 1/8” blade
Wire stripper

## 4. Unpack Reader

Remove Decoder Module box, Hardware and Accessories box and Reader Enclosure from the shipping box. Take Reader Enclosure from bag and flip upside down. Remove two 11/32” flanged nuts (save the nuts) and lift Baseplate from Hood. Open Hardware and Accessories box and remove the hardware bag, then place the 2 flanged nuts in the bag for safekeeping. Depending on your order, there may also be a power supply and spare BA-440-Verifier-Output module, or decals, so carefully open all cartons in the shipping box.



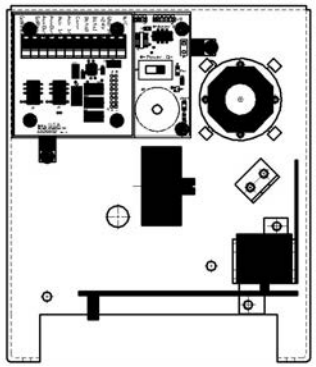
Nuts on bottom of reader



Side view of baseplate



Hardware box

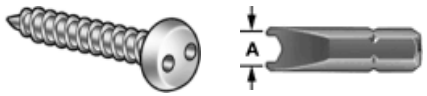


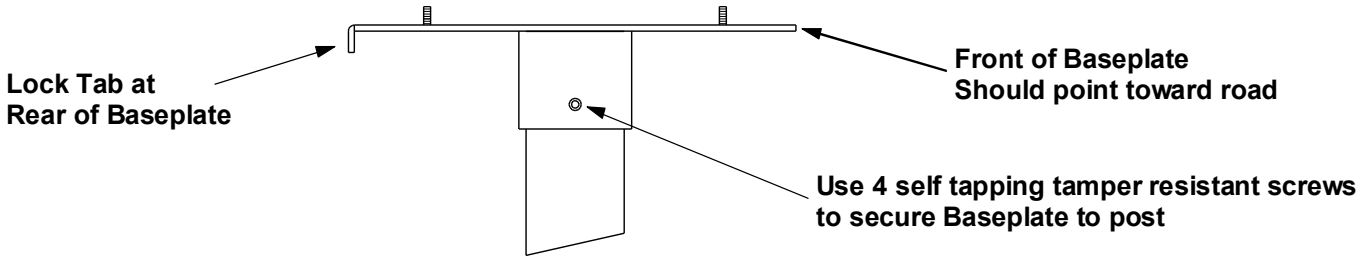
Decoder module

# BA-440 DualBeam Barcode Reader Quickstart Guide Ver 5

## 5. Mount reader Baseplate to mounting post

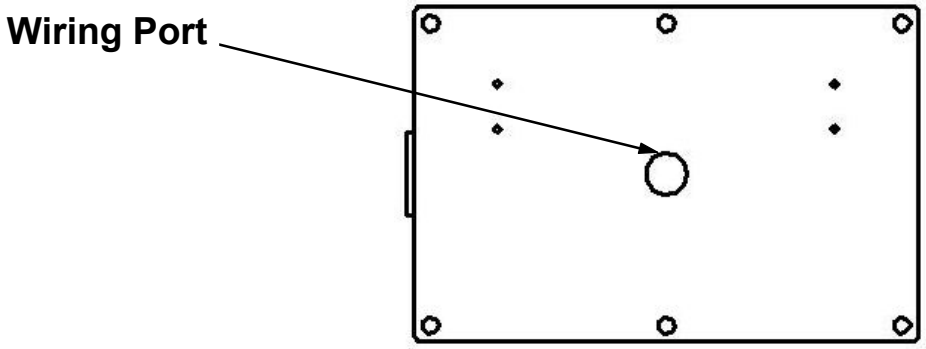
Place the reader Baseplate over the top of the mounting post and thread wiring up through the center hole. The lock tab is on the rear of the Baseplate, be sure the lock tab faces AWAY from the road. Use four (4) Pan head self tapping security screws from the Hardware bag to secure the baseplate to the mounting post. You will need to use the enclosed bit to tighten the tamper resistant screws.




## 6. Seal wiring port in Baseplate

Once the wiring is run up the mounting post and out through the hole in the Baseplate, you can use the 4” putty roll from the hardware bag to seal the opening in the baseplate for wiring. Peel putty from backing and mold around wiring to seal the hole. It is best to have a watertight seal on this opening, BAI recommends using silicone caulk or expanding foam if available. **This step is extremely important; if the wiring port is not sealed the outdoor environment will cause the reader to deteriorate.**

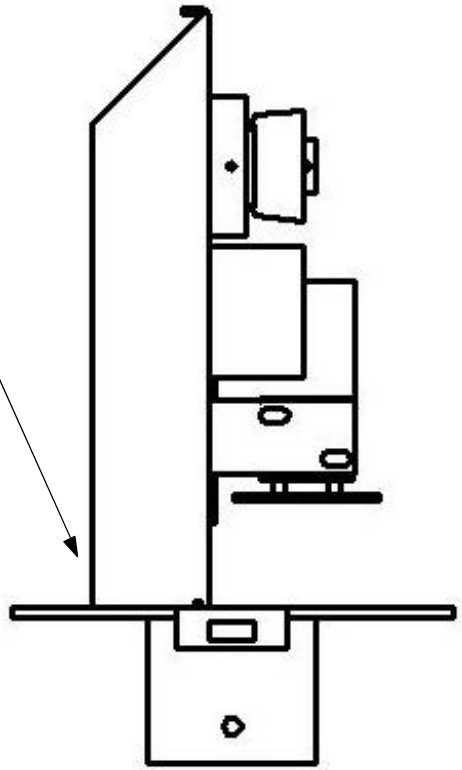


# BA-440 DualBeam Barcode Reader Quickstart Guide Ver 5

## 7. Mount Decoder on Baseplate

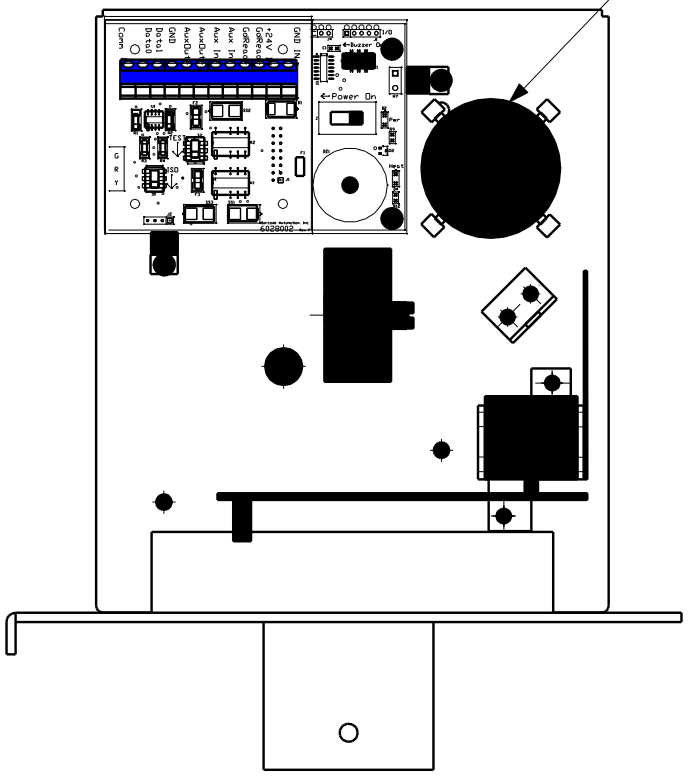
Unpack and set decoder module on Baseplate as shown with the 4 threaded studs passing through the holes on the decoder. Take care not to get fingerprints or dirt on the optics, particularly the gold polygon. This gold polygon should face the road with the terminal block for connections at the rear of the Baseplate. Use the four 5/16” nylock nuts on the threaded studs to secure the decoder module in place. Recommend using a 10” socket extension with the 5/16” socket to tighten the nylock nuts.

**Use four 5/16” nylock nuts to secure decoder to baseplate**



**Rear View Lock Tab at Rear of Baseplate**

**Gold polygon faces road**

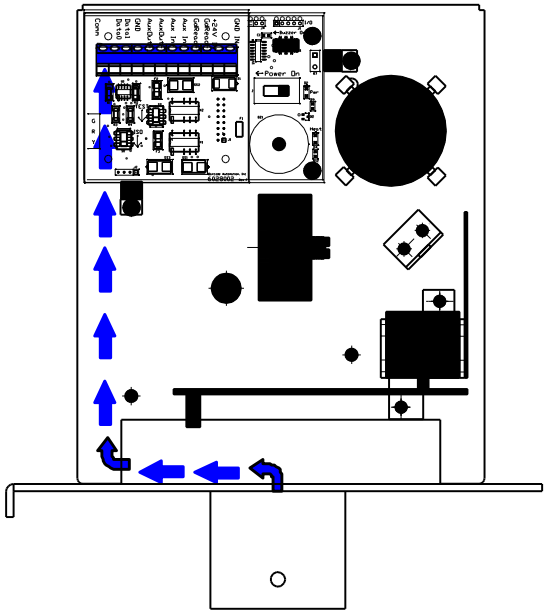


**Right Side View of Baseplate with decoder in place**

# BA-440 DualBeam Barcode Reader Quickstart Guide Ver 5

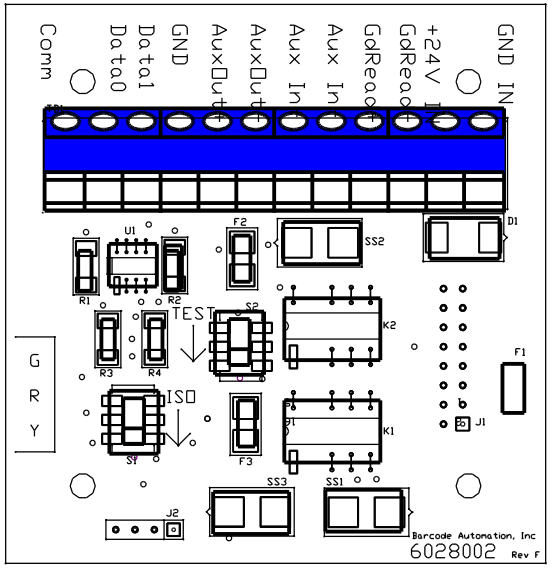
## 8. Route Wiring to Terminals

Take wiring and route it to the terminal block as shown in this Illustration. Take care not to leave wires too long or put stress on the reader circuit boards and components.



## 9. Make Wiring Connections

Strip insulation and connect wiring to Decoder at the terminal block in the upper left hand corner.




| | |
|---|---|
| GND IN | **POWER INPUT** Ground or common input from 24VDC source |
| +24V | **POWER INPUT** Positive input from 24VDC source |
| GdRead+ | **RELAY OUTPUT**, Gate, normally open (NO) |
| GdRead- | **RELAY OUTPUT**, Gate, normally open (NO) |
| Aux In+ | **INPUT** for Vehicle presence trigger |
| Aux In- | **INPUT** for Vehicle presence trigger |
| AuxOut+ | **RELAY OUTPUT**, Auxiliary, normally open (NO) |
| AuxOut- | **RELAY OUTPUT**, Auxiliary, normally open (NO) |
| GND | **GROUND** from reader power input |
| Data1 | **WIEGAND 1** (Data 1) communication output to access control system |
| Data0 | **WIEGAND 0** (Data 0) communication output to access control system |
| Comm | **GROUND** or Common signal connection to access control system |

Recommended wire for power is 18 AWG copper (minimum) for runs up to 200’.

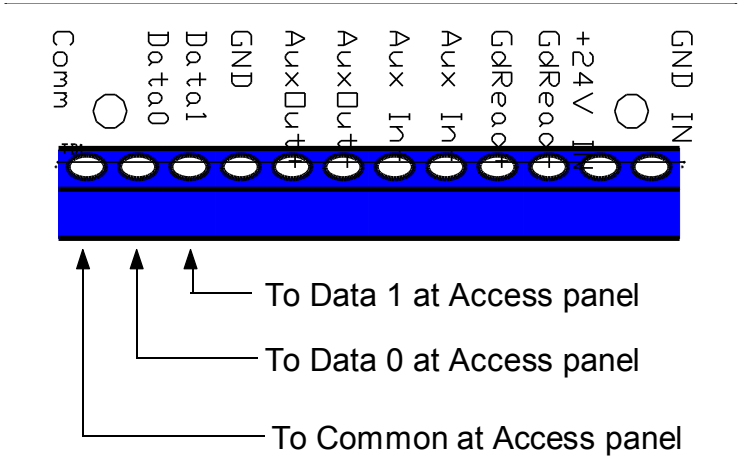
# BA-440 DualBeam Barcode Reader Quickstart Guide Ver 5

## 10. Isolated Wiegand communication

**Optoisolated Connection (Preferred)**

The Wiegand output communication terminals are marked as Data1, Data0 and Common. The normal configuration for Wiegand communication from the BA-440 is optoisolated, where the reader does not supply voltage on the Data0 and Data1 terminals and Comm does not have a direct connection to the reader ground. Voltage and ground reference is supplied by the access panel or telephone entry system. Use 18Ga shielded copper cable for runs up to 400' and ground the shield at the access system end only.

Once Wiegand communication is connected to the access system, **turn the access system on** and measure voltage between Common and Data0, then Common and Data1. There should be approximately 5Vdc present on the Data0 and Data1 terminals. If the voltage is present communications should operate fine.




## 11. Convert to Direct Wiegand connection if Optoisolated connection does not function

**Direct Connection** (if Optoisolated connection cannot be used)

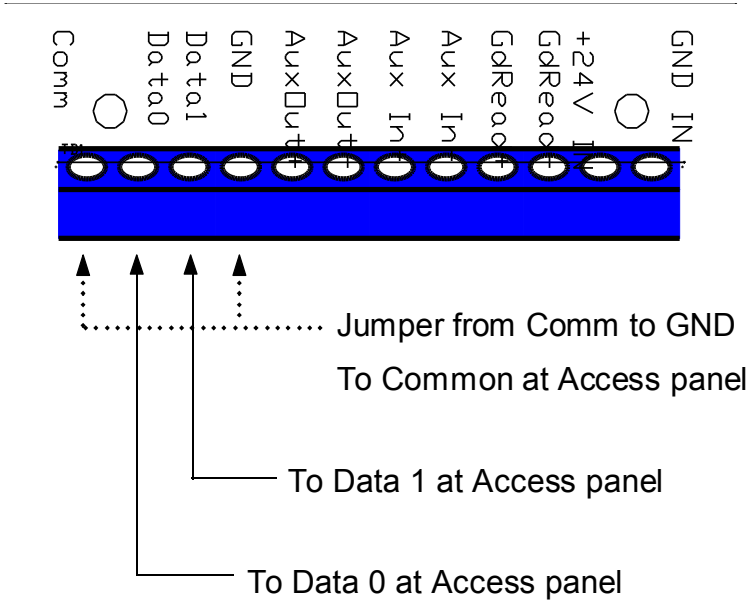
In some cases the Optoisolated Wiegand communication cannot be used, generally due to the access panel or telephone entry not providing voltage on the Data0 and Data1 lines.

To change the Wiegand connection from optoisolated to direct:

1. **Move switch S1 to upper position (away from ISO)**
2. **Place small jumper wire between Comm and GND terminals**

This causes the BA-440 to provide +5Vdc voltage on the Data0 and Data1 lines and connects the Common reference to the reader Ground.

If you have questions about connecting the Wiegand Interface contact BAI at 800-528-9167.

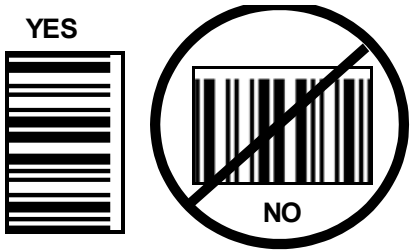


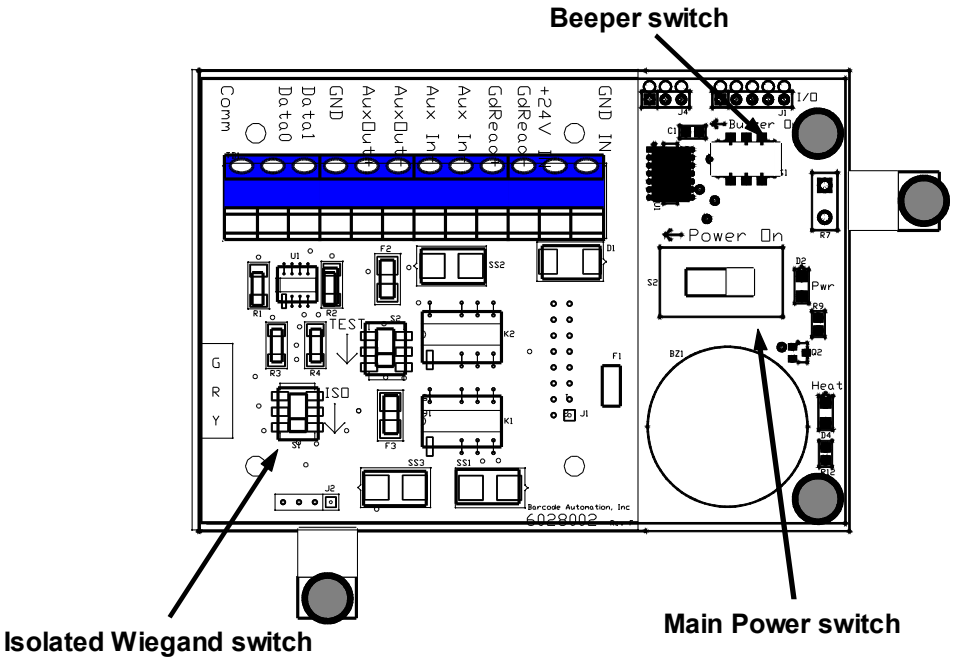
# BA-440 DualBeam Barcode Reader Quickstart Guide Ver 5

## 12. Turn Reader On

Turn Main Power switch On (refer to illustration for location) . You will hear a short beep and the output relays click as the gold polygon begins to spin. Within 15 seconds the laser will turn on and the BA-440 will be ready to operate. Use the enclosed sample barcode or a decal from the site for testing.

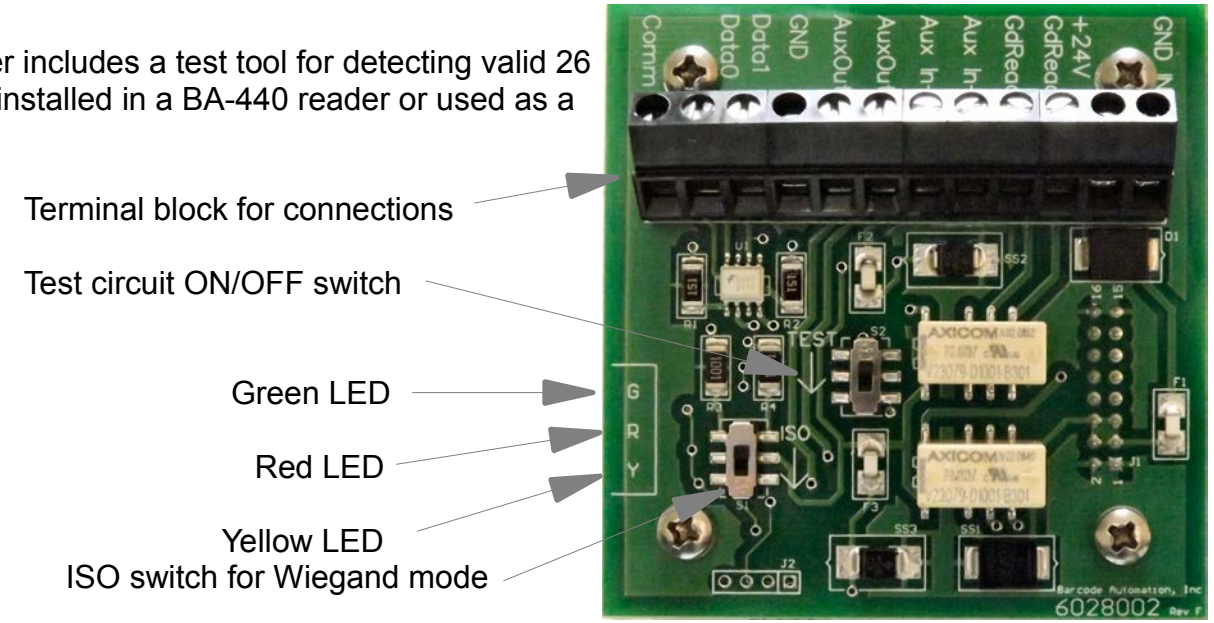
Hold the decal as shown and pass it through the laser lines at least two feet away from the reader. A successful read results in a short beep from the BA-440 and data transmission out both the serial port and Wiegand port. Decal should read out to at least 6' out from the reader. **NOTE: You must wait at least 1 second after a read before the BA-440 will read the same barcode decal again.**




## 13. Test Wiegand output with Verifier

The BA-440-Verifier-Output for the model BA-440 reader includes a test tool for detecting valid 26 bit Wiegand messages. The Verifier can be used when installed in a BA-440 reader or used as a handheld Wiegand message verifier.



# BA-440 DualBeam Barcode Reader Quickstart Guide Ver 5

## 13. Test Wiegand output with Verifier, continued




The Green, Red, and Yellow LED indicators are mounted on the rear of the circuit board. Look **behind** the left side of the Verifier for the LED indicators as shown here.

The LED's are located in this position in an attempt to keep them out of direct sunlight. You may still need to shade them with your hand if the Sun shines directly on them.

To use the Wiegand verifier function with the module mounted on a BA-440, turn the "test" switch ON, which is down toward the arrow. The Green, Red, and Yellow LED's on the left side of the board will blink **once** in sequence to show that the verifier is ready. Remember that the LED's are **behind** the circuit board as shown, you should stand behind the reader to see them clearly.

**NOTE:** To preserve battery life, the verifier circuit will run for 10 minutes after the switch is ON, then it will go into power saving mode (hibernation). To reactivate the verifier, turn the TEST switch OFF, wait for 5 seconds, then ON again. The 3 LED's will blink **once** in sequence to show that the circuit is live.

Once the TEST switch is ON and you have seen the LED's blink, watch the Verifier LED's when a decal is read:

No LED's blinking -

1. Check the voltage on Data 0 and Data 1 to see if there is 5 VDC present. If no voltage is found or if voltage is less than 3 Vdc, Wiegand messages cannot be sent. Be sure the access system is connected and turned ON.
2. If voltage measures 5 Vdc, check to see if Wiegand messages are being sent by the reader by observing the Data0 and Data1 LED's on the Decoder circuit board. If these are not blinking, no messages are being sent.
3. If Data 0 and Data1 LED's on the reader are blinking but the Verifier does not then there is no message output from the reader to detect.

Green LED Only blinking - This indicates a 26 bit Wiegand message is detected with valid parity.

Amber LED Only blinking - This indicates a 26 bit Wiegand message is detected with bad parity. Usually this means that the Data 0 and Data 1 connections are reversed, so try reversing the Data 0 and Data 1 leads on the Verifier and see if the Green LED blinks.

Red LED Only blinking -

1. This indicates that the Verifier is detecting less than 26 bits or more than 26 bits in a group. Since this is not a valid message the Red LED will blink. Check voltages on Data 0 and Data 1 - if either one is 0 or less than 3 Vdc that may be the problem.
2. If red blinks occur when no Wiegand message is sent by the reader there may be electrical interference on the communication line. This interference may be caused by a power line too close to the communications cable or a disconnected shield.

# BA-440 DualBeam Barcode Reader Quickstart Guide Ver 5

## 13. Test Wiegand output with Verifier, continued

Red and Green blinking - This indicates that both valid messages and electrical interference are detected on the communications line at the same time. Some Wiegand messages will fail if the interference occurs when the message is transmitted.

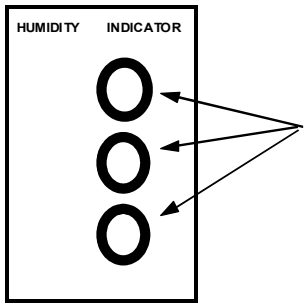
As an added feature, the Verifier circuit is battery powered and can be used as a handheld Wiegand 26 bit verifier for any equipment using 26 bit Wiegand communication. To use the module as a handheld device, Turn the ISO switch down toward the arrow, then connect the Data 0, Data 1, and Common terminals to the Wiegand communication line you want to verify and turn the test circuit ON.

Remember that the test circuit is live for 10 minutes after you turn the test switch ON, then it will go into hibernation. This is to preserve battery life if the technician forgets to turn the test circuit off. To restart the test circuit, turn the test switch OFF, wait 5 seconds, then turn back ON. You should see the Green, Red, and Yellow LED's blink **once** in sequence to show that the circuit is live.

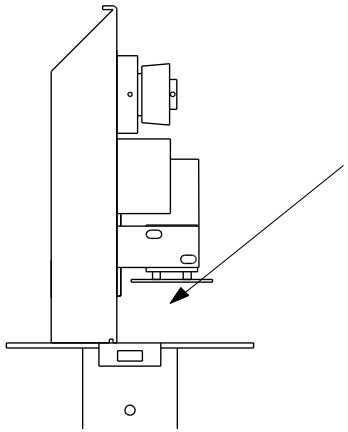
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## 14. Place Desiccant pack on Baseplate

Open a Desiccant pouch, then place both bags and Humidity card on Baseplate under the receiver circuit board as shown. Desiccant should be changed every 6 months during regular optical cleaning. Every time a Model 440 Reader Hood is removed check the humidity card. If all circles are pink the desiccant must be replaced before the reader Hood is put back on.




If all circles are pink replace desiccant and humidity card.



**Place Desiccant bag and Humidity card here**

**Rear View**
**Lock Tab at**
**Rear of Baseplate**

# BA-440 DualBeam Barcode Reader Quickstart Guide Ver 5

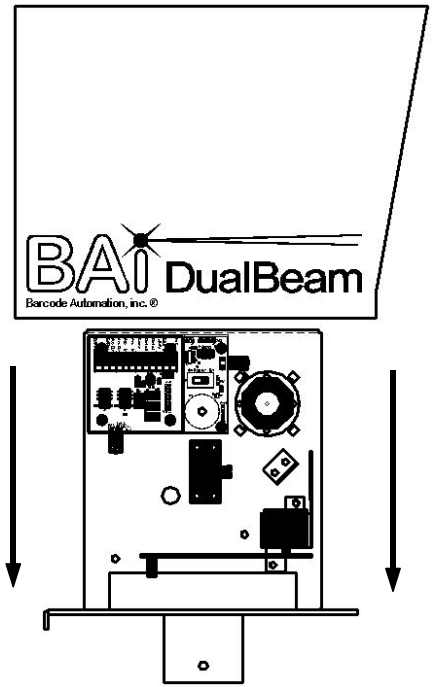
## 14. Install and seal the reader Hood

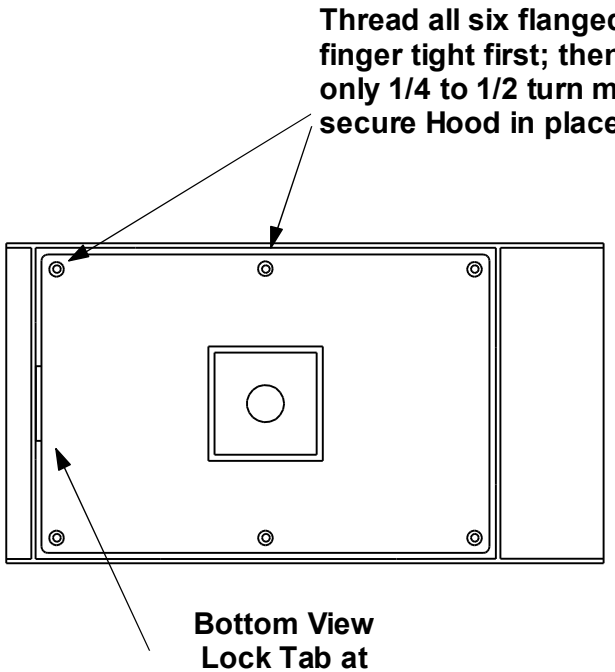
Set the Hood in place over the top of the BA-440 decoder assembly so that the six studs pass into the six holes on the baseplate. The front window of the Hood goes by the gold polygon as shown. Secure the Hood using the six 11/32” flanged nuts from the hardware bag. The Hood must be secured using all 6 flanged nuts for a good seal.

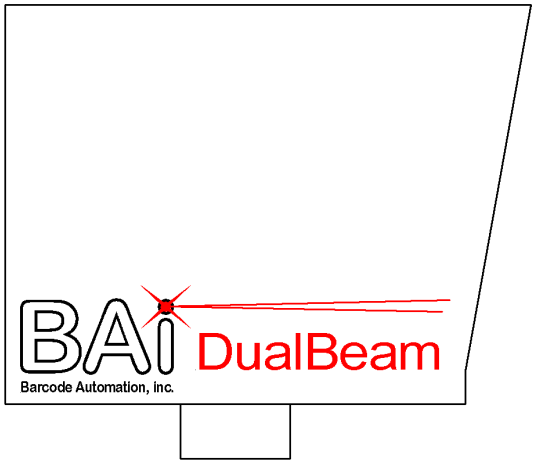
**IMPORTANT: Thread all 6 nuts finger tight first, then tighten only 1/4 to 1/2 turn more on each nut.**

**CAUTION: The seal on the reader Hood and hardware may be damaged if nuts are overtightened.**

To prevent unauthorized access a padlock (not included) may be placed through lock tab at the rear of the BA-440.




**Assembly complete.**
**Contact BAI at 800-528-9167 with any questions.**